

Graphs

- Graph is a discrete structure with vertices and edges connecting the vertices.
- A graph $G = (V, E)$ consists of V , a nonempty set of vertices (or nodes) and E , a set of edges. Each edge has either one or two vertices associated with it, called its endpoints. An edge is said to connected its endpoints.

Applications of Graph

- Graphs are used to solve problems in many fields. Some of them are as follows:
- Graphs are used to model the geographic maps of cities in which each place in city can be represented by the node and the road connecting such places are represents the edge
- Graphs are used to model the computer network in which each node is a machine and link between them represent the edge
- They are used to analyze the electric circuits, project planning, genetics etc
- Any structured problem can be modeled by graphs. Then can help to solve typical problems those connected with finding shortest path of most economical route between two vertices, or the smallest set of edges which connect all the vertices in a graph
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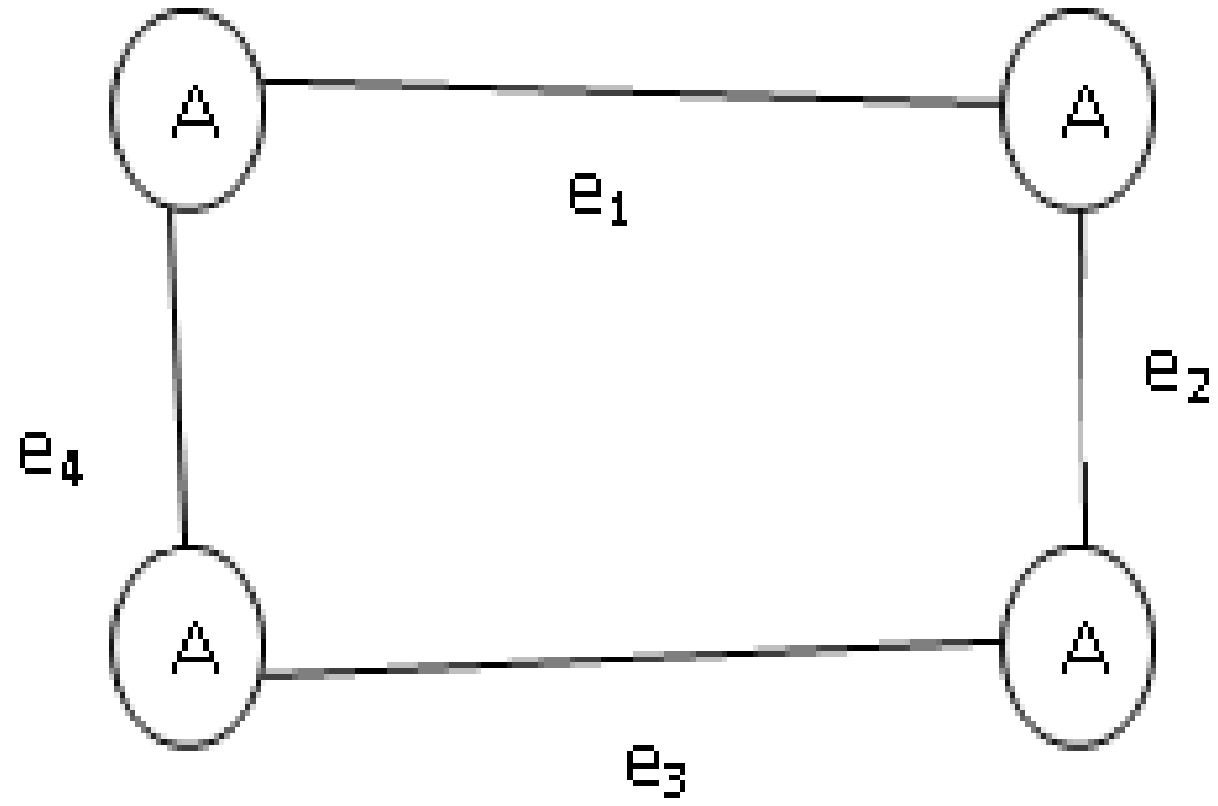
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- Graphs are used to study the structure of WWW
- It is used to find the number of different combination of flight between two cities in an airline network
- It can be used to distinguish between two chemical compounds with the same molecular formula but different structure
- It is also used to assign channels to television stations

Types of Graphs

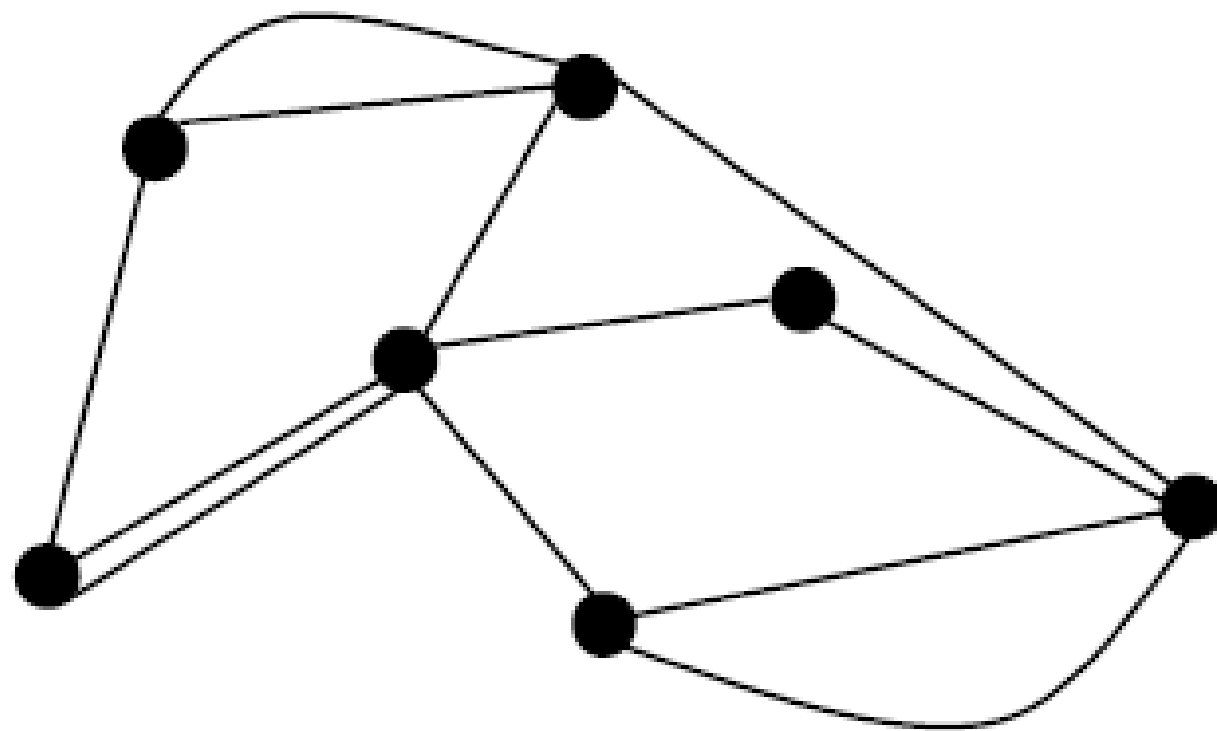
- **Simple Graph**

- We define a simple graph as 2 – tuple consists of a non empty set of vertices V and a set of unordered pairs of distinct elements of vertices called edges.
- We can represent graph as $G = (V, E)$.
- This kind of graph has no loops and can be used for modeling networks that do not have connection to themselves but have both ways connection when two vertices are connected but no two vertices have more than one connection.
- The figure below is an example of simple graph.



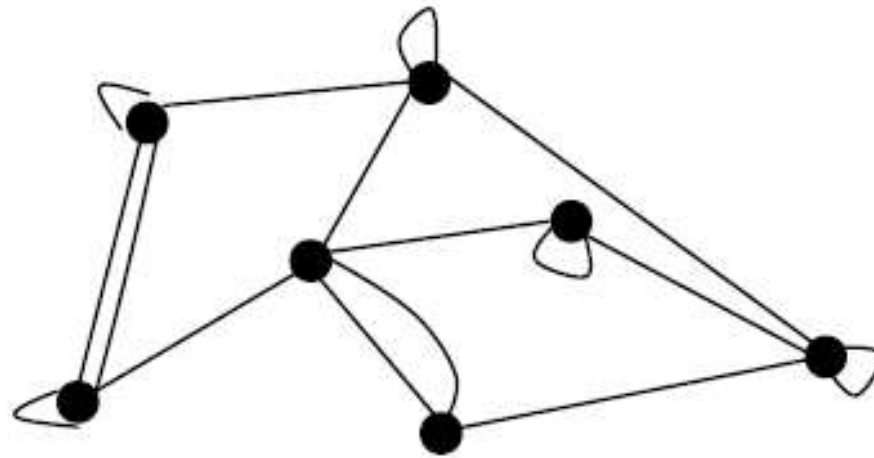
Multigraph

- A multigraph $G = (V, E)$ consists of a set of vertices V , a set of edges E , and a function f from E to $\{\{u, v\} \mid u, v \in V, u \neq v\}$.
- The edges e_1 and e_2 are called multiple or parallel edges if $f(e_1) = f(e_2)$.
- In this representation of graph also loops are not allowed.
- Since simple graph has single edges every simple graph is a multigraph.
- The figure below is an example of a multigraph.



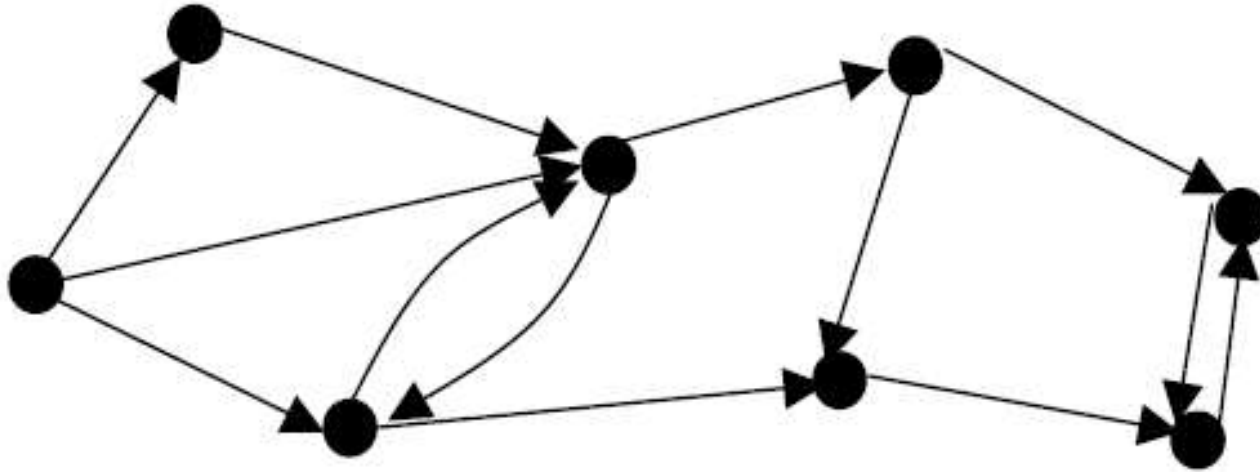
Pseudograph

A pseudograph $G=(V, E)$ consists of a set of vertices V , a set of edges E , and a function f from E to $\{\{u, v\} | u, v \in V\}$. An edge is a loop if $f(e) = \{u, u\} = \{u\}$ for some $u \in V$. The figure below is an example of a multigraph.



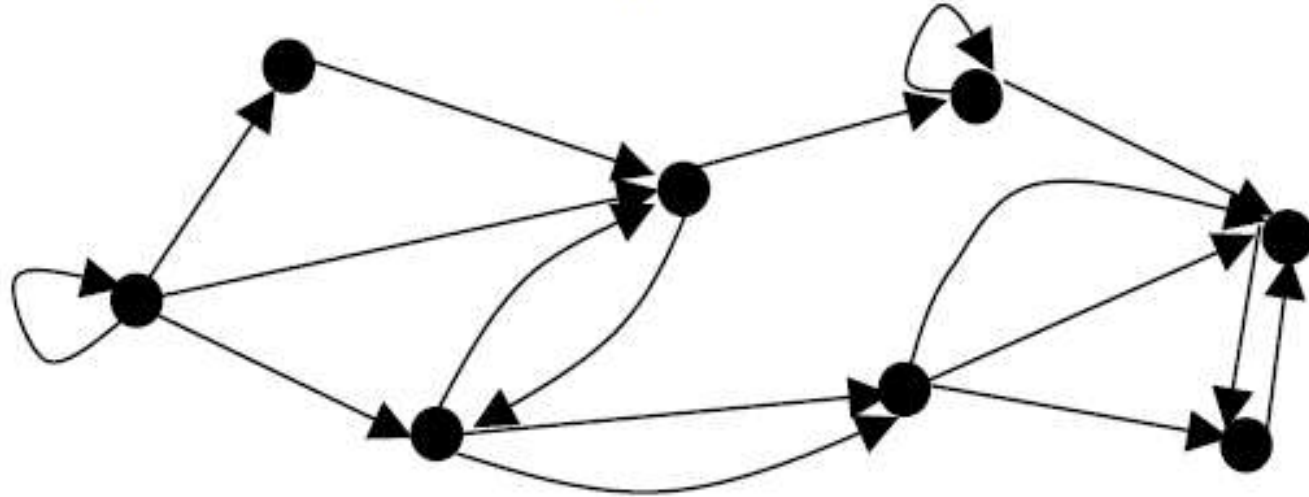
Directed Graph

A directed graph (V, E) consists of a set V of vertices, a set E of edges that are ordered pairs of elements of V . The below figure is a directed graph. In this graph loop is allowed but no two vertices can have multiple edges in same direction.



Directed Multigraph

A directed multigraph $G=(V, E)$ consists of a set of vertices V , a set of edges E , and a function f from E to $\{(u, v)|u, v \in V\}$. The edges e_1 and e_2 are called multiple edges if $f(e_1) = f(e_2)$. The figure below is an example of a directed multigraph.



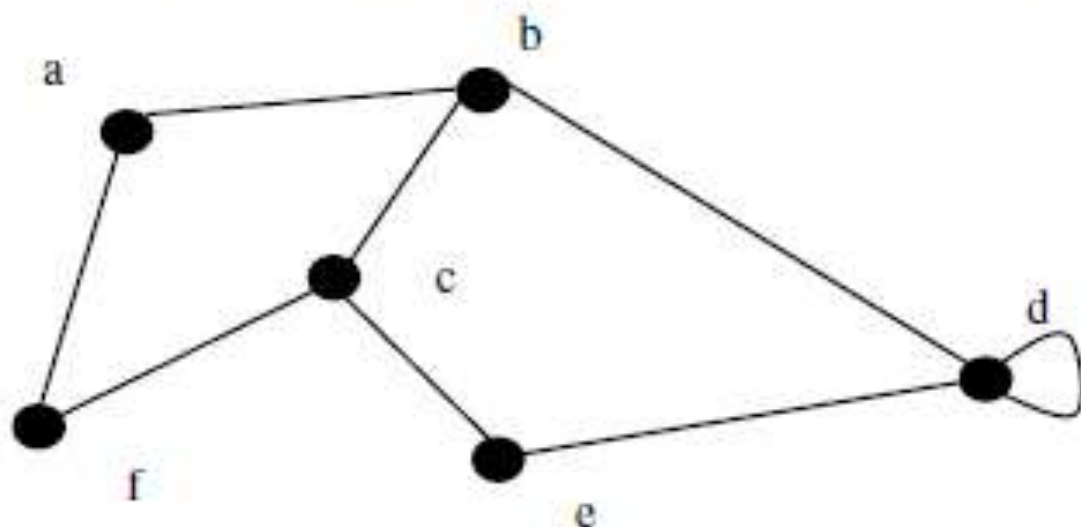
Terminologies

Two vertices u, v are adjacent vertices of a graph if $\{u, v\}$ is an edge.

The edge e is called incident with the vertices u and v if $e = \{u, v\}$. This edge is also said to connect u and v , where u and v are end points of the edge.

Degree of a vertex in an undirected graph is the number of edges incident with it, except a loop at a vertex. Loop in a vertex counts twice to the degree. Degree of a vertex v is denoted by $\deg(v)$. A vertex of degree zero is called isolated vertex and a vertex with degree one is called pendant vertex.

Example: Find the degrees of the vertices in the following graph.



Solution:

$$\deg(a) = \deg(f) = \deg(e) = 2 ; \deg(b) = \deg(c) = 3; \deg(d) = 4$$

Theorem 1: The Handshaking Theorem

Let $G = (V, E)$ be an undirected graph with e edges. Then $2e = \sum_{v \in V} \deg(v)$.

Theorem 2:

An undirected graph has an even number of vertices of odd degree.

Proof:

Take two sets of vertices, V_1 , a set of vertices with even degree, and V_2 , a set of vertices with odd degree. In an undirected graph $G=(V, E)$ we have

$$2e = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

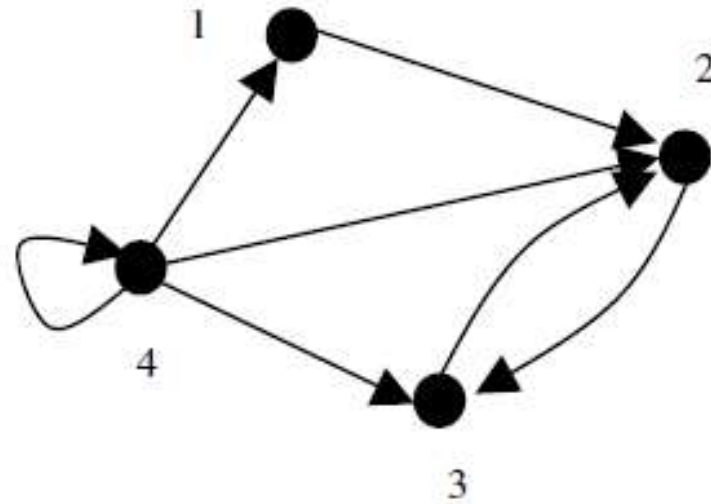
From the equality above we can say the left part is even i.e. $2e$ is even, the sum of $\deg(v)$ for $v \in V_1$ is even since every vertices has even degree. So for the left hand to be even sum of $\deg(v)$ for $v \in V_2$ must be even. Since all the vertices in the set V_2 have odd degree the number of such vertices must be even for the sum to be even. Hence proved.

Let (u, v) be an edge representing edge of a directed graph G . u is called adjacent to v and v is called adjacent from u . The vertex u is called **initial vertex** and the vertex v is called **terminal or end vertex**. Loop has same initial and terminal vertex.

In directed graph the **in-degree** of a vertex v , denoted by $\deg^-(v)$, is the number of edges that have v as their terminal vertex. The **out-degree** of a vertex v , denoted by $\deg^+(v)$, is the number of edges that have v as their initial vertex. Loop at a vertex adds up both in-degree and out-degree to one more than calculated in-degree and out-degree.

Example:

Find the in-degree and out-degree of each vertex in the following graph.

**Solution:**

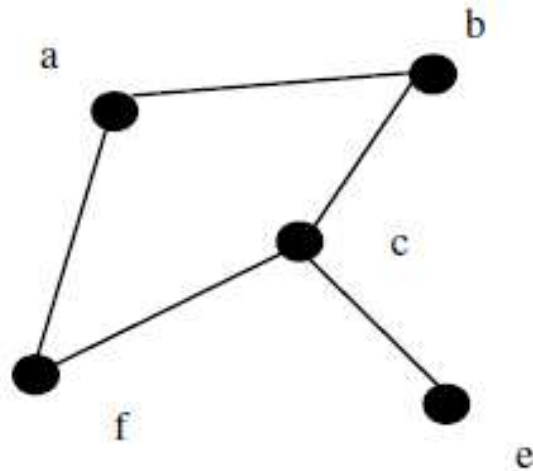
In-degrees of a graph are $\deg^-(1) = \deg^-(4) = 1$; $\deg^-(2) = 3$; $\deg^-(3) = 2$ and the out-degrees of a graph are $\deg^+(1) = \deg^+(2) = \deg^+(3) = 1$; $\deg^+(4) = 4$.

Theorem 3:

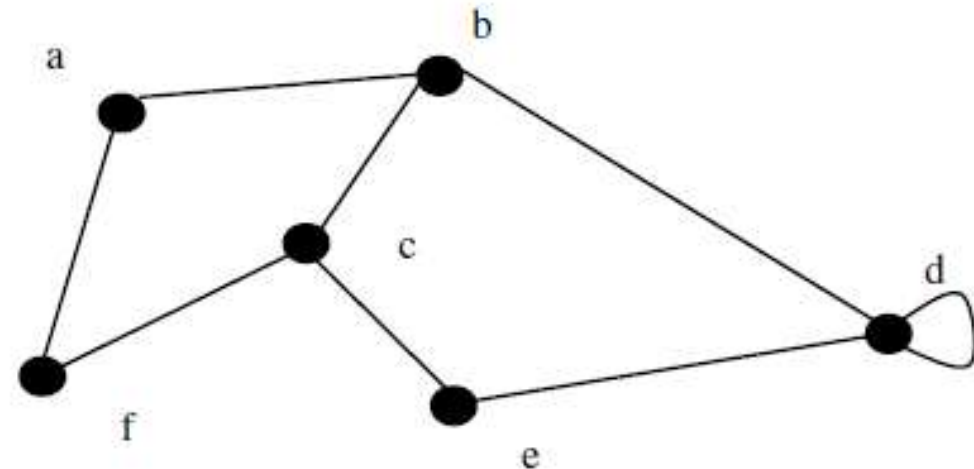
Let $G(V, E)$ be a graph with directed edges. Then $\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$.

A **subgraph** of a graph $G = (V, E)$ is a graph $H = (W, F)$ where $W \subseteq V$ and $F \subseteq E$.

Example:



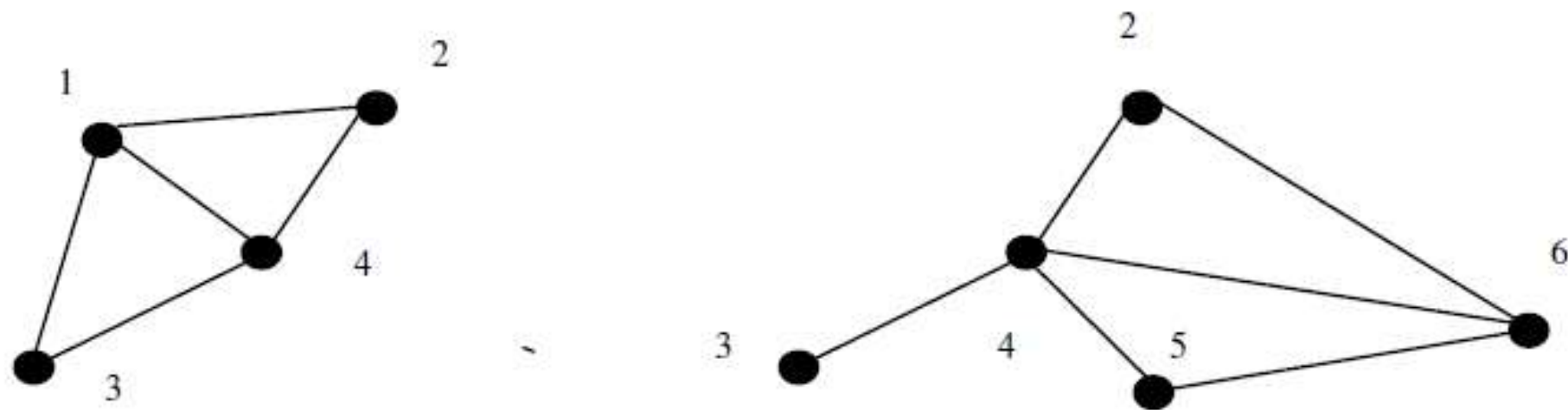
is a subgraph of



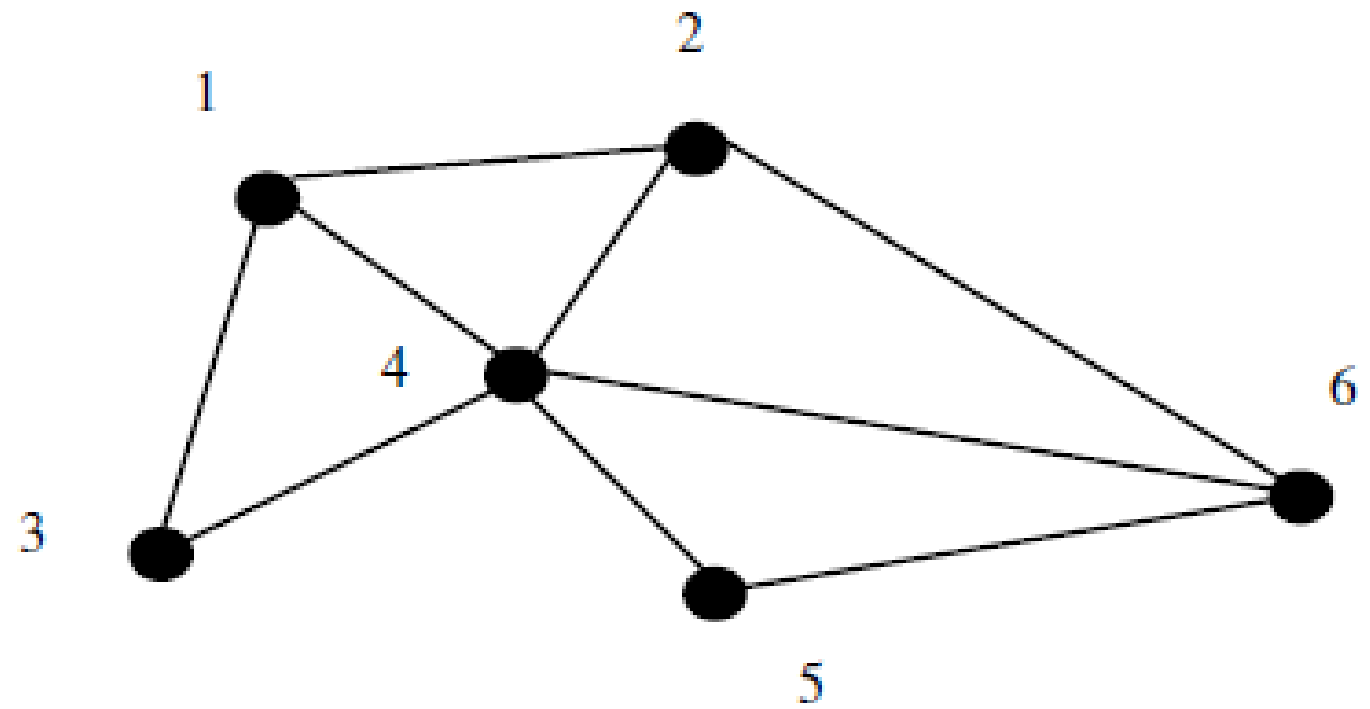
The **union** of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with vertex set $V_1 \cup V_2$ and the edge set $E_1 \cup E_2$. The union of G_1 and G_2 is denoted by $G_1 \cup G_2$.

Example:

Find the union of two graphs given below.



Solution:



Complete Graphs

The complete graph of n vertices, denoted by K_n , is the simple graph that contains exactly one edge between each pair of distinct vertices.

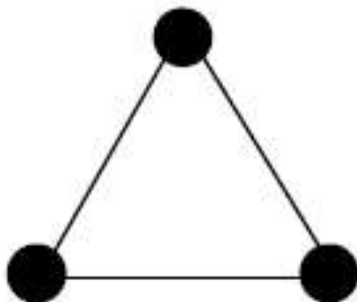
Example:

What are K_1 , K_3 , and K_5 ?

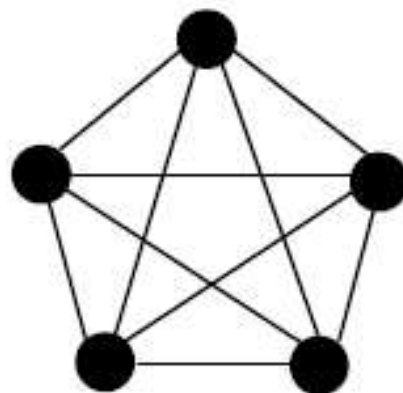
Solution:



K_1



K_3



K_5

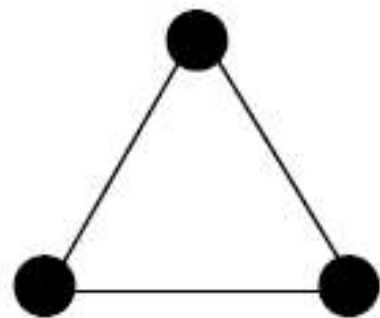
Cycles

The cycle C_n , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.

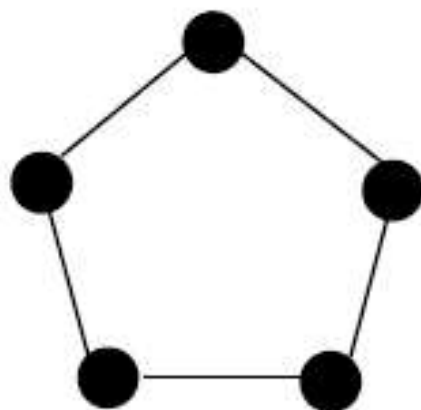
Example:

What are C_3 , C_5 , and C_7 ?

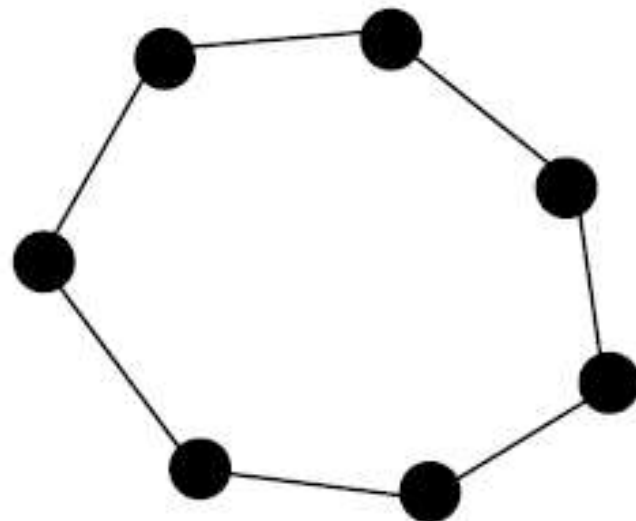
Solution:



C_3



C_5



C_7

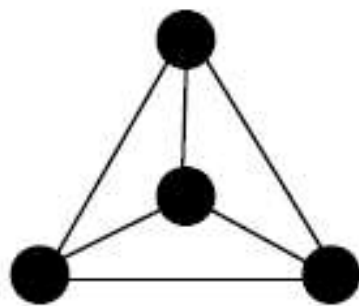
Wheels

The wheel W_n , for $n \geq 3$, is an union of C_n and additional vertex where the new vertex is connected by each vertex of the cycle.

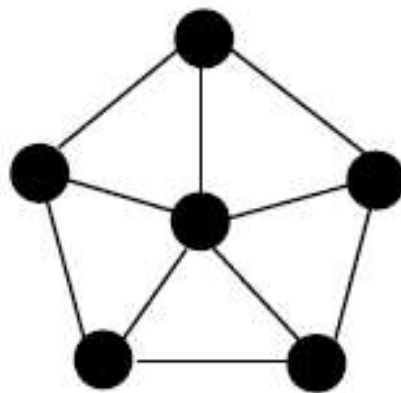
Example:

What are C_3 , C_5 , and C_7 ?

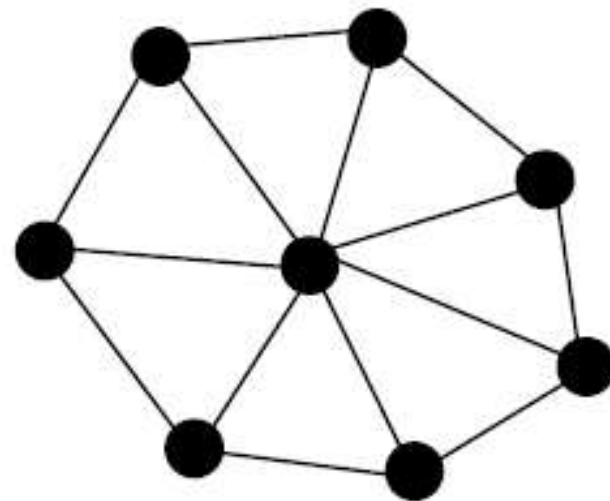
Solution:



W_3



W_5



W_7

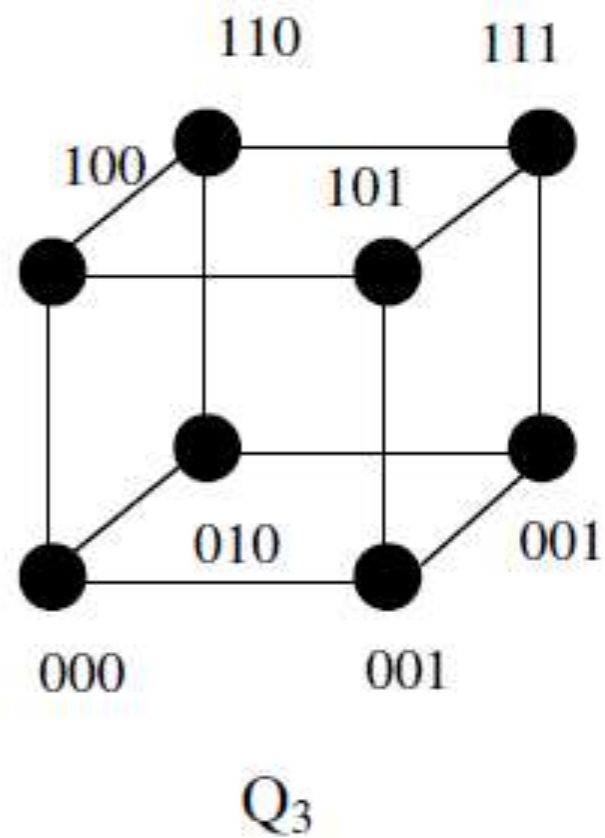
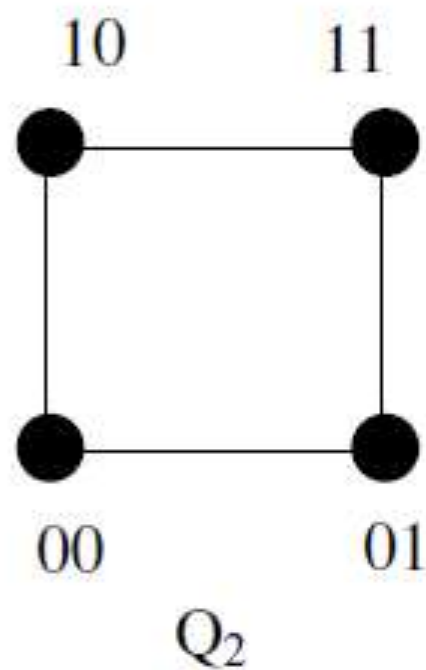
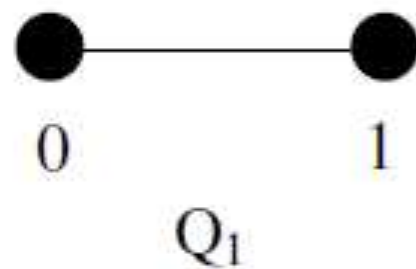
n- Cubes

The n -dimensional cube, or n -cube, denoted by Q_n , is the graph that has vertices representing the 2^n bit strings of length n . Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.

Example:

What are Q_1 , Q_2 , and Q_3 ?

Solution:



Bipartite Graphs

A simple graph G is bipartite if its vertex set V can be partitioned into two disjoint subsets V_1 and V_2 such that every edge in the graph connects a vertex from the set V_1 to the vertex of the set V_2 . No two vertices of the same set are connected by an edge.

A graph is bipartite if and only if it is possible to color the vertices of the graph with at most two colors such that no two adjacent vertices have the same color.

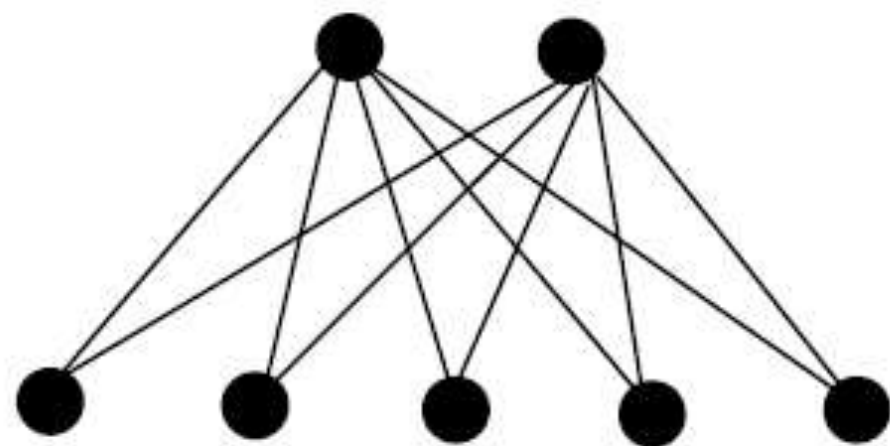
A graph is bipartite if and only if it is not possible to start at a vertex and return to this vertex by traversing an odd number of distinct edges.

Complete Bipartite Graphs: The complete bipartite graph $K_{m,n}$ is the graph where the vertex set is partitioned into two subsets of m and n vertices, respectively. In this graph there is an edge between two vertices if and only if two vertices are in different subsets of vertices.

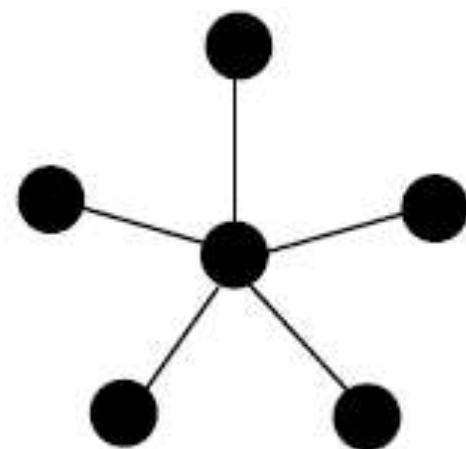
Example:

Sketch $K_{2,5}$, $K_{1,5}$.

Solution:



$K_{2,5}$



$K_{1,5}$

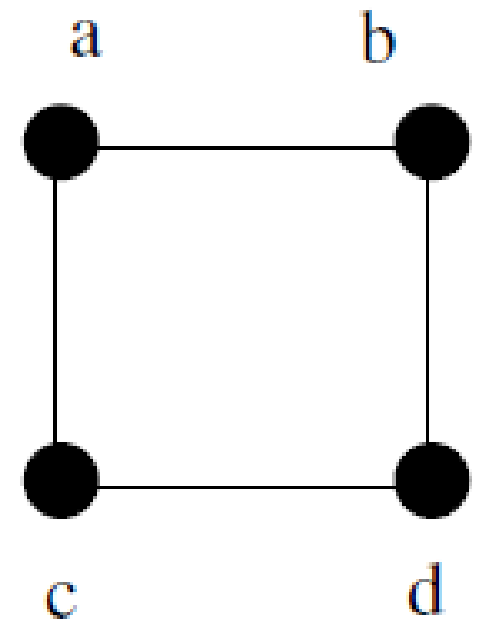
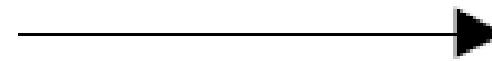
Graph Representations

Graph can be represented in many ways; one of the ways of representing a graph without multiple edges is by listing its edges. Some other ways are described below:

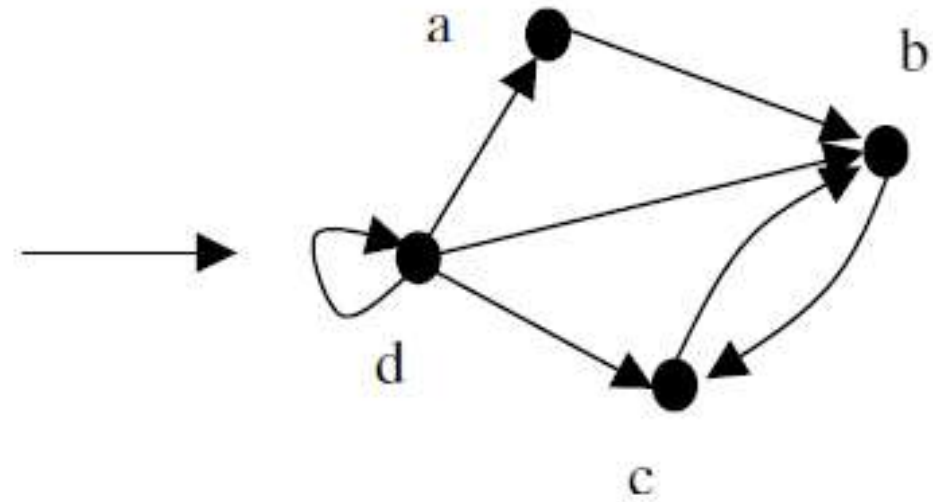
Adjacency List

This type of representation is suitable for the undirected graphs without multiple edges, and directed graphs. This representation looks as in the tables below.

Edge List for Simple Graph	
Vertex	Adjacent Vertices
a	b, c
b	a, d
c	a, d
d	b, c



Edge List for Directed Graph	
Initial Vertex	End Vertices
a	b
b	c
c	b
d	a, b, c, d



If we try to apply the algorithms of graph using the representation of graphs by lists of edges, or adjacency lists it can be tedious and time taking if there are high number of edges. For the sake of the computation, the graphs with many edges can be represented in other ways. In this class we discuss two ways of representing graphs in form of matrix.

Adjacency Matrix

Given a simple graph $G = (V, E)$ with $|V| = n$. assume that the vertices of the graph are listed in some arbitrary order like $v_1, v_2, \dots, v_n$. The adjacency matrix A of G , with respect to the order of the vertices is n -by- n zero-one matrix ($A = [a_{ij}]$) with the

condition, $a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$. Since there are n vertices and we may

order vertices in any order there are $n!$ possible order of the vertices. The adjacency matrix depends on the order of the vertices, hence there are $n!$ possible adjacency matrices for a graph with n vertices.

Adjacency matrix for undirected graph is symmetric, in case of the pseudograph or multigraph the representation is similar but the matrix here is not zero-one matrix rather the $(i, j)^{\text{th}}$ entry of the matrix contains the number of edges appearing between that pair of vertices.

In case of the directed graph we can extend the same concept as in undirected graph as dictated by the relation $a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$. The main difference is that the matrix may not be symmetric.

If the number of edges is few then the adjacency matrix becomes sparse. Sometimes it will be beneficial to represent graph with adjacency list in such a condition.

Incidence Matrix

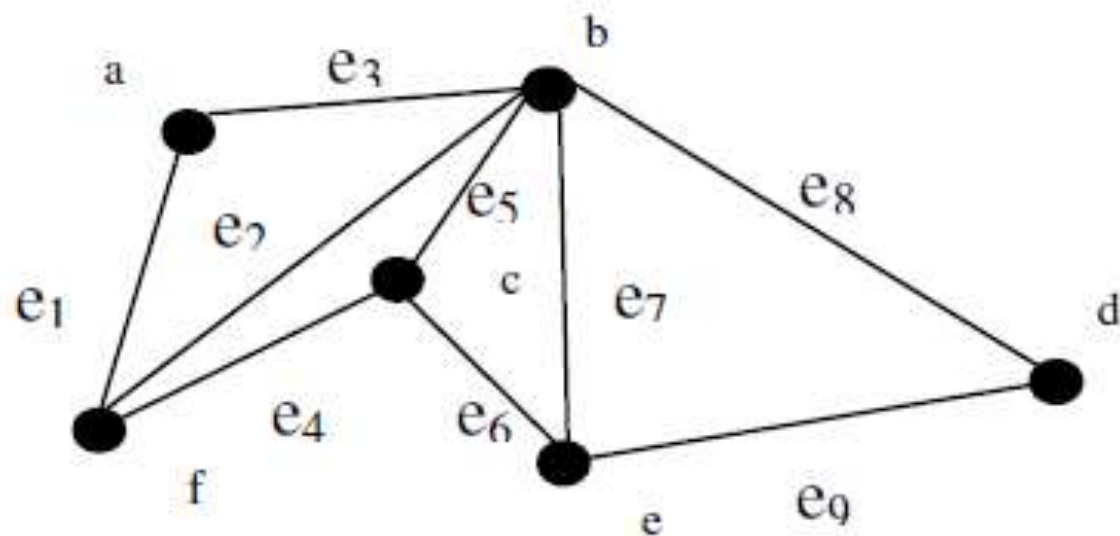
This is another way of representing graph. Given an undirected graph $G = (V, E)$. Assume that the vertices of the graph are $v_1, v_2, \dots, v_n$ and the edges of the graph are $e_1, e_2, \dots, e_m$. the incidence matrix of a graph with respect to the above ordering of V and E

is n -by- m matrix $M = [m_{ij}]$, where $m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$.

When the graph is not simple then also the graph can be represented by using incidence matrix where multiple edges corresponds to two different columns with exactly same entries. Loops are represented with column with only one entry.

Example #1:

Represent the following graph using adjacency matrix and incidence matrix.



Solution:

Let the order of the vertices be a, b, c, d, e, f and edges order be $e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9$.

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

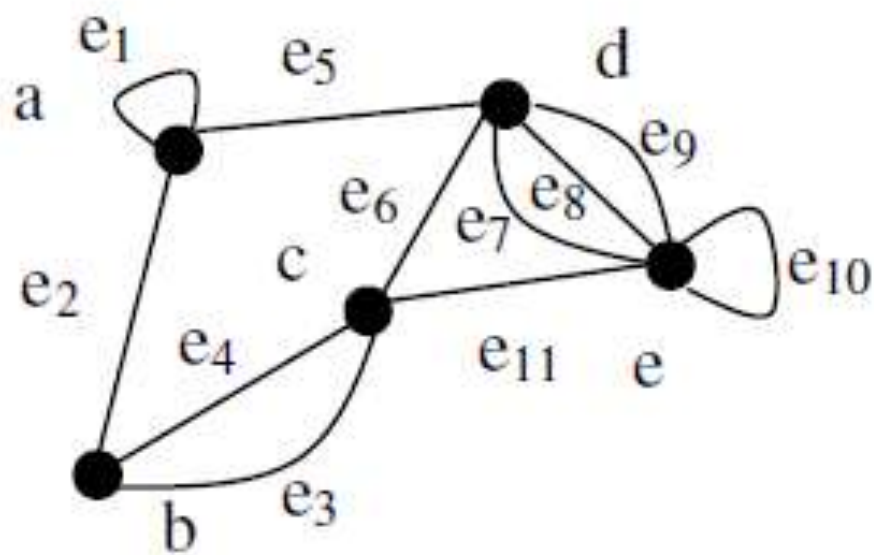
Adjacency Matrix

$$\begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} \begin{bmatrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 & e_9 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Incidence Matrix

Example #2:

Represent the following graph using adjacency matrix and incidence matrix.



Solution:

Let the order of the vertices be a, b, c, d, e, f and edges order be $e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9, e_{10}, e_{11}$.

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 3 & 1 \end{bmatrix}$$

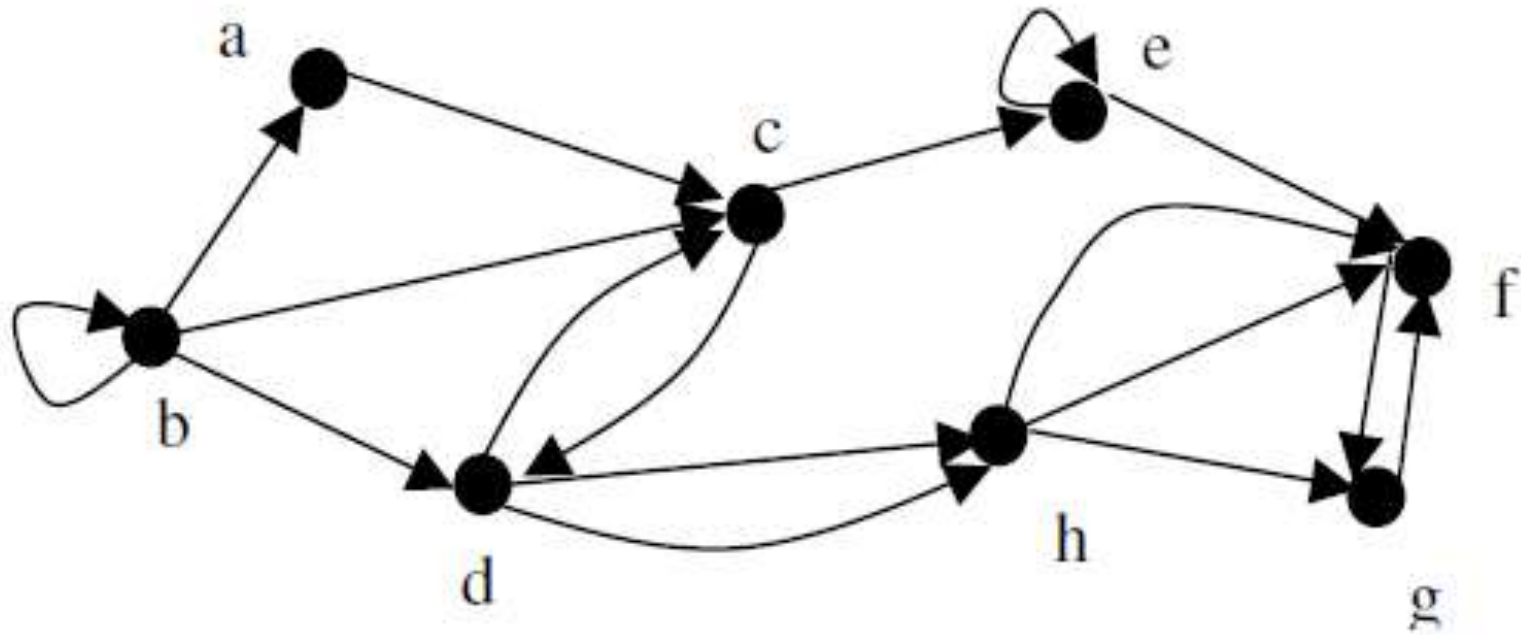
Adjacency Matrix

$$\begin{array}{c} a \\ b \\ c \\ d \\ e \end{array} \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 & e_9 & e_{10} & e_{11} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

Incidence Matrix

Example #3:

Represent the following directed graph using adjacency matrix.



Solution:

Let the order of the vertices be a, b, c, d, e, f, g and h. The adjacency matrix is

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \end{bmatrix}$$

Graph coloring

A coloring of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

A chromatic number of a graph is the least number of colors needed for a coloring of this graph.

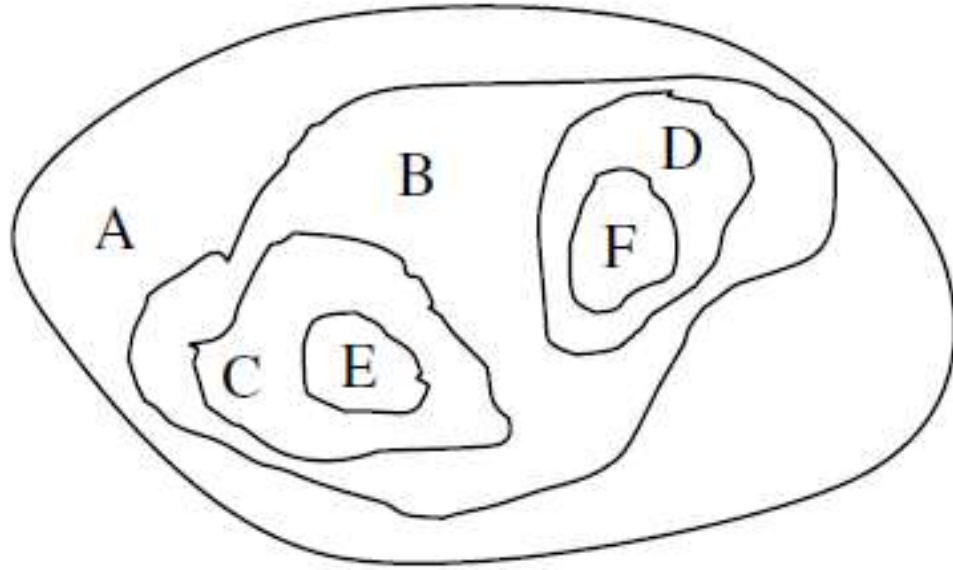
Theorem 11: The Four Color Theorem

The chromatic number of a planar graph is no greater than four.

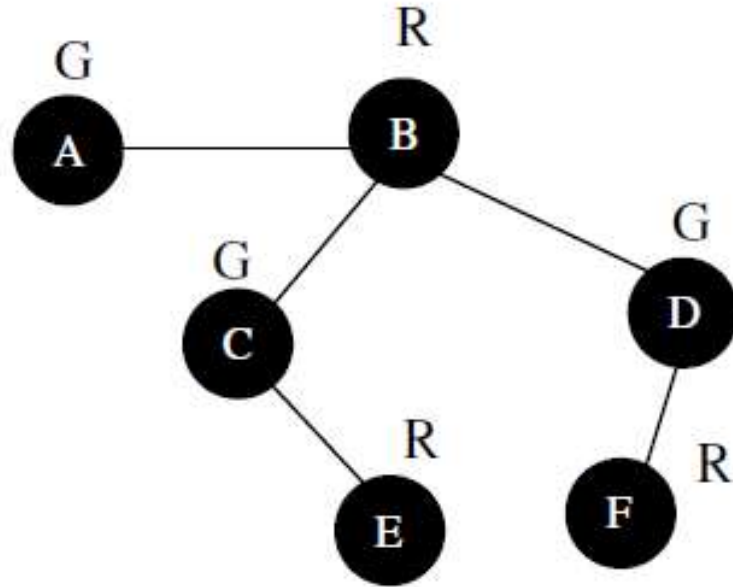
To show the chromatic number of a graph as k we must show that the graph can be colored using k colors and the given graph cannot be colored using fewer than k colors.

Example#1

Construct the dual graph for the map shown. Then find the number of colors needed to color the map so that no two adjacent regions have the same color.



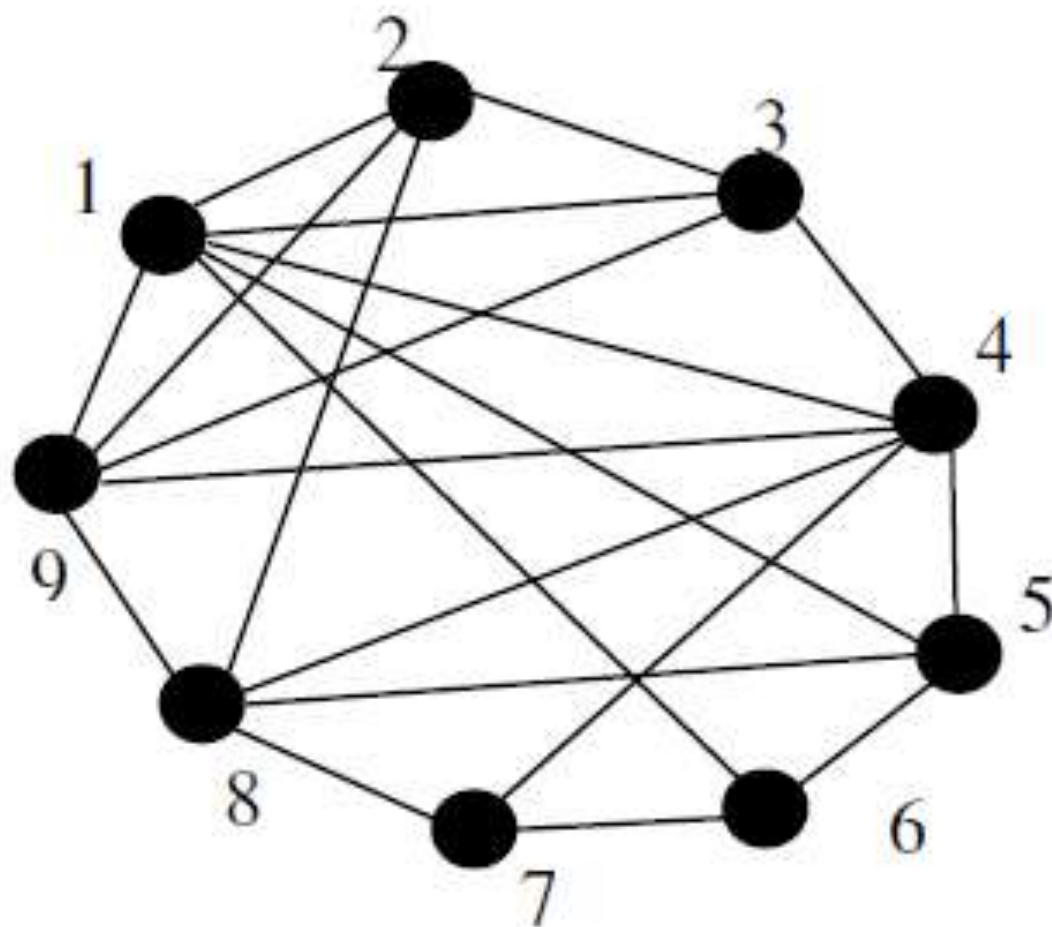
Solution:



We can color the graph with at most 2 colors as shown in the graph. Take R and G as red and green

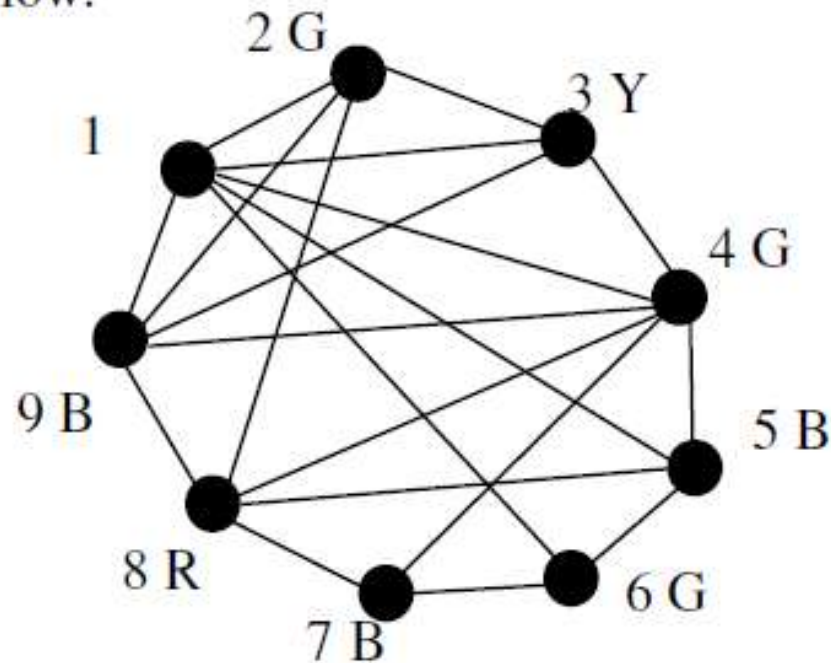
Example#2

Find the chromatic number of the graph below.



Solution:

Lets start with vertex 1, it has adjacent vertices as 2, 3, 4, 5, 6, 9 so using only 2 colors would suffice for the graph having the edges from 1 to its adjacent vertices. However since 9 has its adjacent vertices as 1, 2, 3, 4 we cannot just color the above graph with 2 colors because at least 1, 2 and 9 must have different colors. Trying with 3 colors we found that at least 1, 2, 3, and 9 must have different colors. So trying with four colors we can color the graph. Hence the chromatic number of the above graph is 4. possible coloring is shown in the figure below.



Hamilton paths

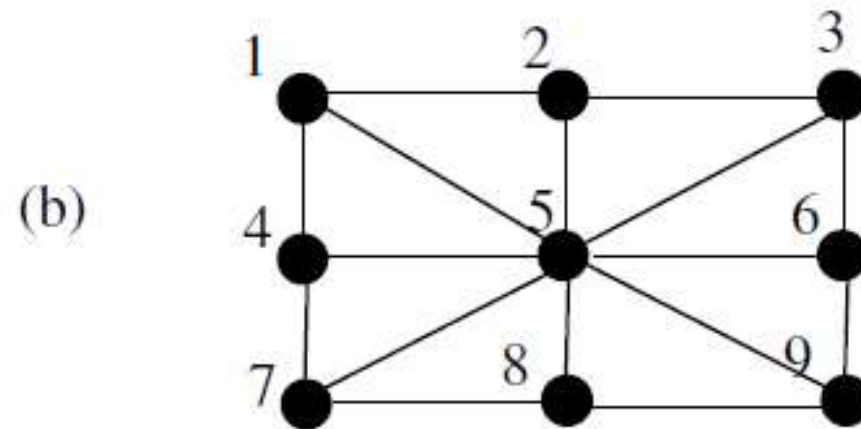
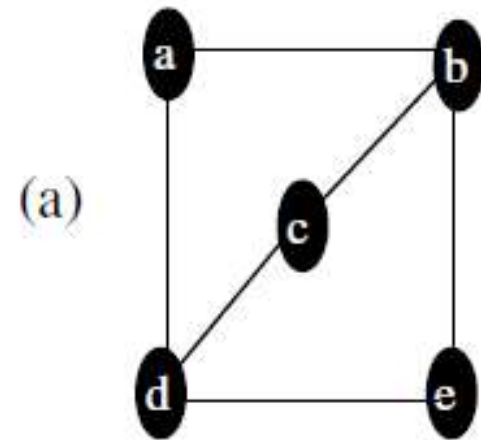
A path $x_0, x_1, \dots, x_{n-1}, x_n$ in the graph $G = (V, E)$ is called Hamilton path if $V = \{x_0, x_1, \dots, x_{n-1}, x_n\}$ and $x_i \neq x_j$ for $0 \leq i < j \leq n$. A circuit $x_0, x_1, \dots, x_{n-1}, x_n, x_0$, with $n > 1$, in a graph $G = (V, E)$ is a Hamilton circuit if $x_0, x_1, \dots, x_{n-1}, x_n$ is a Hamilton path.

A graph with a vertex of degree one cannot have a Hamilton circuit.

If a vertex in a graph has degree 2, then both edges incident with this vertex must be part of Hamilton cycle.

Example:

Find Hamilton circuit from the following graph if exists? What about Hamilton path?



Solution:

In graph (a) there is no Hamilton circuit since the node c has degree 2 and both the edges from it must be in Hamilton circuit, which is not possible. One of the Hamilton path in the graph (a) is a, b, c, d, e.

In graph (b) we can find Hamilton circuit, the circuit can be 1,2,3,5,6, 9,8,7,4,1. Since there is circuit we can have path also.